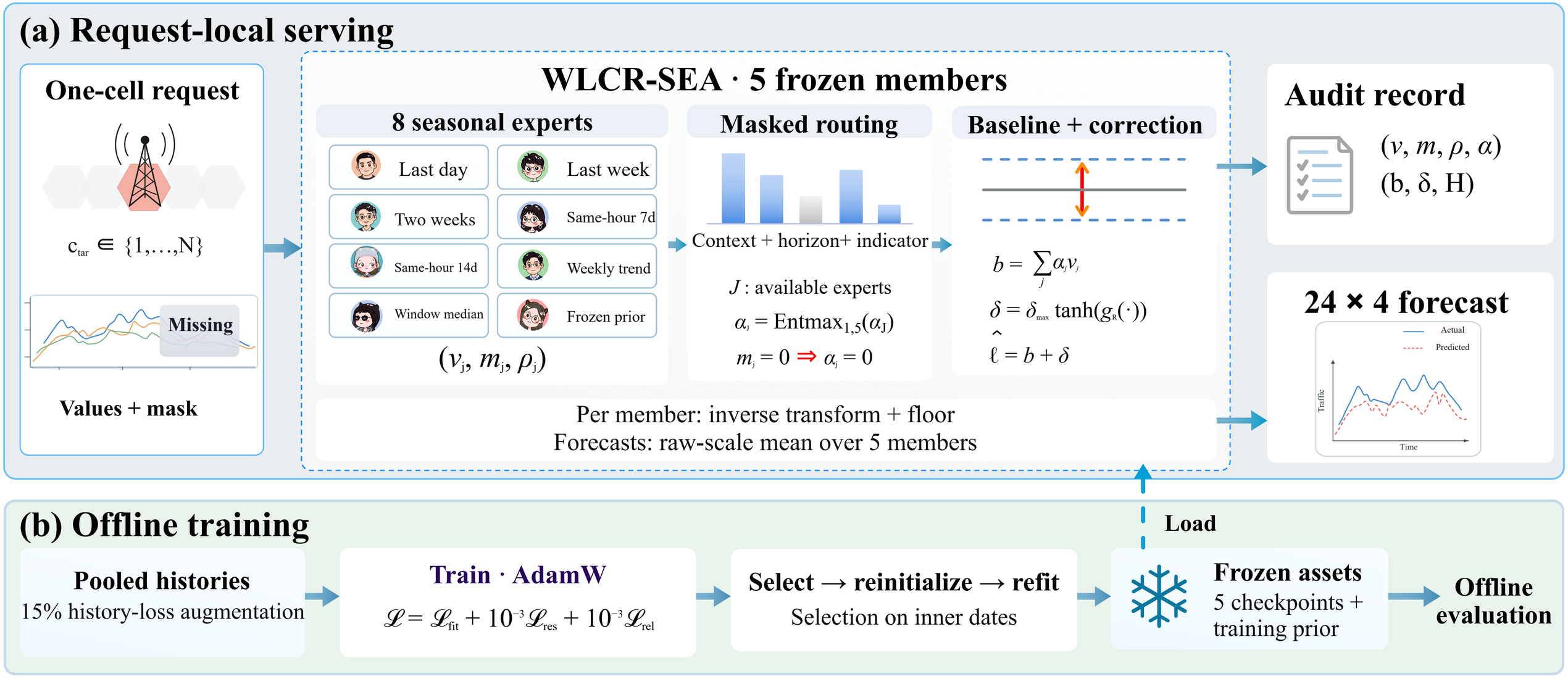

(a) Request-local serving
One-cell request
$c_{tar} \in \{1,\ldots,N\}$
Missing
Values + mask
WLCR-SEA · 5 frozen members
8 seasonal experts
Last day
Last week
Two weeks
Same-hour 7d
Same-hour 14d
Weekly trend
Window median
Frozen prior
(v_j, m_j, ρ_j)
Masked routing
Context + horizon+ indicator
J : available experts
$\alpha_J = \mathrm{Entmax}_{1,5}(\alpha_J)$
$m_j = 0 \Rightarrow \alpha_j = 0$
Baseline + correction
$b = \sum_j \alpha_j v_j$
$\delta = \delta_{max} \tanh(g_R(\cdot))$
$\hat{\ell} = b + \delta$
Per member: inverse transform + floor
Forecasts: raw-scale mean over 5 members
Audit record
(v, m, ρ, α)
(b, δ, H)
24 × 4 forecast
Actual
Predicted
Traffic
Time
(b) Offline training
Pooled histories
15% history-loss augmentation
Train · AdamW
$\mathscr{L} = \mathscr{L}_{fit} + 10^{-3}\mathscr{L}_{res} + 10^{-3}\mathscr{L}_{rel}$
Select → reinitialize → refit
Selection on inner dates
Load
Frozen assets
5 checkpoints + training prior
Offline evaluation